

Description of Roman CPIP Data Challenge 2

I. INTRODUCTION

Here we describe the plan for CPIP Data Challenge 2. Our goal is to study and compare analysis choices by running simulated cosmic shear analyses with data vectors measured from weak lensing mocks. Subsequent subchallenges with simulated 3x2pt data vectors will follow at some point. The idea of the main challenge is to make available a set of 25 blind data vectors generated with different values for cosmological parameters and systematics, specifically with a diversity of baryonic feedback scenarios. The participants should take the approach of their choice for cosmological inference and for dealing with small scales and attempt to recover those values. A simulated covariance matrix will also be made available. We will be thus verifying the impact on constraints that come from particular analysis choices. In this sense, the scope of this challenge differs from that of DC1 mainly in that (1) realistic data vectors will be used to prepare for the actual analysis of the medium tier survey, and (2) the focus will be on comparing different analysis choices and modeling strategies from the large set of options currently available in Cocoa instead of requiring all users to use the same main pipeline configurations.

II. DESCRIPTION OF THE MOCKS

A. N-body simulations

The Gower Street v2 simulations are a set of N-body cosmological simulations generated through the PKDGRAV3 code, and presented as a set of lightcone snapshots over 100 shells, which range from $z=0$ to $z=49$.

The suite is composed of two different setups, each with their own simulations. The **small** setup uses a simulation box size of $L=500\text{Mpc}/h$ and a number of particles of $N=1000^3$, with the minimum halo mass of $10^{11}M_\odot$. The **big** setup has a larger box size of $L=1.25\text{Gpc}/h$, and it has $N=1350^3$ particles, with a higher mass threshold for haloes (minimum halo mass of $10^{12}M_\odot$). The output of both setups includes a halo catalog for each simulation. Additionally, the number count of halos is presented in a per-pixel basis on NSIDE 4096 Healpix maps, one for each redshift shell.

The simulation set is separated between runs at a fiducial cosmology and varying cosmology runs. At a fiducial ΛCDM cosmology, it includes 200 simulations with the **small** setup and 200 simulations at the **big** setup. On varying cosmology, with the **small** setup, we have 600 simulations with ΛCDM and 1200 simulations with $w\text{CDM}$. Additionally, with the **big** setup, we have 850 simulations at ΛCDM cosmologies, 2500 simulations at $w\text{CDM}$ cosmologies, and 1000 simulations at varying

w_0w_a cosmologies. The w_0w_a set is currently being expanded and new simulations are being added to it.

B. Weak lensing maps

We use a forward modeling approach to generate weak lensing mocks from the Gower St v2 lightcone simulations. For each simulation realization, we start with the corresponding cosmological parameters, and can add to it a set of nuisance parameters, which include multiplicative shear bias corrections m_i , photometric redshift shifts δz_i , source clustering bias amplitudes b_i^{sc} , and NLA intrinsic alignment amplitude A_{IA} and redshift dependence η_{IA} .

We start by generating a set of mocks from the fiducial **big** simulation boxes, to be used for the computation of the simulated covariance matrix. For this run, all multiplicative and additive bias terms are set to zero, and intrinsic alignment is disabled ($A_{\text{IA}} = 0$). The fiducial lightcone realizations have $\Omega_m = 0.3176$, $\sigma_8 = 0.8253$, $\Omega_b = 0.049$, $h = 0.67$, $n_s = 0.96$, and $w_0 = -1$.

We consider a scenario with eight tomographic bins, with $n(z)$ distributions corresponding to those of the fiducial medium-tier scenario, as described in Section III. These distributions are rebinned and cross-matched with the Gower Street lightcone shell edges.

C. Forward modeled systematics

For each tomographic bin, the photometric redshift shift Δz_i is applied by shifting the $n(z)$ prior to projection, and the resulting per-shell galaxy weights $n_{\text{gal}}^{(i)}(z_j)$ are used throughout the remainder of the pipeline.

The projected matter overdensity $\delta(\hat{n}, z_j)$ in each lightcone shell j is loaded from the simulation output as a **HEALPix** map and downgraded to **NSIDE** = 2048 resolution for faster computation and less storage usage of the final mocks. Optionally, baryonic effects can be applied to the density shells using the **BaryonForge** package, which displaces particles following the selected baryonic correction model. For the fiducial run presented here, baryonification is disabled.

From the projected density shells, the weak lensing convergence κ , shear (γ_1, γ_2) , and intrinsic alignment fields are computed via the Limber approximation. The intrinsic alignment contribution follows the non-linear alignment (NLA) model, with density weighting applied implicitly through source clustering.

The total effective shear in each tomographic bin i is then accumulated over all shells as

$$\gamma_{1,2;i}^{\text{tot}}(\hat{n}) = \sum_j n_{\text{gal}}^{(i)}(z_j) [\gamma_{1,2}(z_j) + \gamma_{1,2}^{\text{IA}}(z_j) A_{\text{IA}}] [1 + b_i^{\text{sc}} \delta(\hat{n}, z_j)], \quad (1)$$

where the source clustering term $b_i^{\text{sc}} \delta$ modulates the effective number of source galaxies correlated with the foreground density field.

We generate an artificial galaxy catalog by sampling pixel positions from the weighted projected density field $d_{\text{tot}}^{(i)}(\hat{n})$, which includes the source clustering modulation.

The full-sky area of $\Omega_{\text{sky}} = 41253 \text{ deg}^2$ is used throughout, to be masked to the Roman medium-tier footprint in post-processing. Each simulated galaxy is assigned intrinsic ellipticity components drawn from a Gaussian distribution with per-component dispersion $\sigma_e = 0.26$. We compute and apply source clustering corrections.

The final output per tomographic bin consists of:

- **g1_map, g2_map**: noiseless shear maps,
- **e1, e2**: shear maps including shape noise,
- **e1n, e2n**: pure noise maps after source-clustering corrections,
- **e1r_map0_ref, e2r_map0_ref**: reference noise maps without source-clustering correction,
- **var_**: per-pixel variance map of the reference noise.

The full-sky mocks are divided into eight maps using the fiducial medium-tier footprint. From this larger set of mocks, we compute simulated $\xi_{\pm}$ covariance matrices, which will be made available for the DC2 participants.

III. REDSHIFT DISTRIBUTIONS

The redshift distributions for the first data challenge were generated by assuming an effective density of $n_{\text{eff}} = 41.3 \text{ arcmin}^{-2}$ distributed across bins proportionally to the integrated $n(z)$ weight. For DC2, we updated our binning to follow a realistic distribution generated from DC25 sims with slimfarmer photometry. The sample is selected to be $S/N > 18$ on YJH mean-combined map. The $n(z)$ is obtained with Roman-sompz, with an effective density per bin of $n_{\text{eff}} = 37.78 \text{ arcmin}^{-2}$. This version is intended to be a more realistic, though still conservative, choice than that of DC1. The distributions are shown in Figures 1 and 2.

IV. CHALLENGE SETUP

For the DC2 data vectors, we will follow the procedure above to generate mocks at different w CDM cosmologies and with systematics. We use **BaryonForge** to

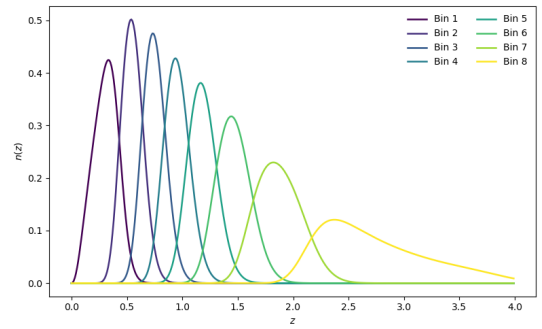


FIG. 1. Redshift distributions for DC1, which follow an effective density of $n_{\text{eff}} = 41.3 \text{ arcmin}^{-2}$ and assume galaxies to be distributed across bins proportionally to the integrated $n(z)$ weight.

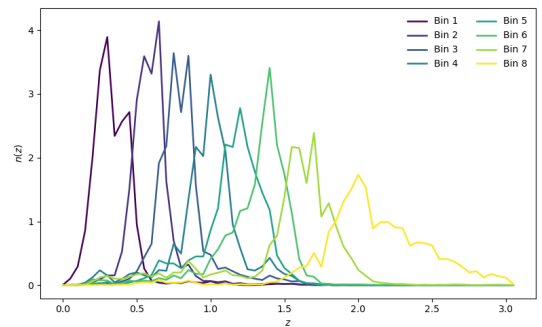


FIG. 2. Redshift distributions for DC2, which follow a realistic distribution generated from DC25 sims with slimfarmer photometry.

apply baryonification following the model described by Anbajagane *et al.* [1]. After generating the baryonified lightcone sims, we choose values for A_{IA} and η_{IA} to run the forward modeling pipeline. We generate a total of 25 mocks each with different w CDM cosmologies and different values for the NLA and baryonic parameters. From each of these mocks, we cut one medium-tier footprint and measure the two-point correlation functions. We use 15 log-spaced θ bins with $\theta_{\text{min}} = 1.5'$ and $\theta_{\text{max}} = 200'$. We will make the data vectors available for the DC2 participants without informing the participants of the true value of the cosmological parameters and of systematics.

As an additional later part of the challenge, we will also generate a few simulations with a more complex IA infusion process, highlighting non-zero tidal torquing.

The main goal for the data challenge participants will be to use our provided mock data vectors and infer the correct cosmological parameters. While DC1 was focused on making sure all participants could run the standard 3x2pt Cocoa pipeline, DC2 aims to make the participants explore more deeply weak lensing inference and find the ideal modeling choices for more realistic data vectors. For that purpose, we are making use of data from mocks and giving freedom for the participants to adopt their model-

ing choices to deal with systematics. Due to our focus on small scales and the availability of three baryonic modeling methods in **Cocoa** (PC, $SP(k)$ and BC_{emu}), The participants are also encouraged to submit, as an extended goal, the estimated power spectrum suppression at small scales. Note that this isn't equivalent to infer precisely the values of all baryonic free parameters, but instead serves as a consistency check between the descrip-

tion used in the theoretical model and the baryonification methodology of the simulations.

V. DIFFSKY MOCK CATALOG

In this section, we will start describing the secondary challenge that will follow the main one and include mocks with source and lens galaxies for 3x2pt analysis.

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- [1] D. Anbajagane, S. Pandey, and C. Chang, Map-level baryonification: Efficient modelling of higher-order correlations in the weak lensing and thermal sunyaev-

zeldovich fields, The Open Journal of Astrophysics **7**, 10.33232/001c.126788 (2024).